ADVANCED QUANTUM MECHANICS

Exercise 14 - 03.02.2026

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Galilean invariance of the Schrödinger equation

We work in one spatial dimension x and time t. The free Schrödinger equation is

$$\mathrm{i}\hbar\,\partial_t\psi(x,t) = -\frac{\hbar^2}{2m}\,\partial_x^2\psi(x,t). \tag{1}$$

A Galilean boost with velocity v is defined by the coordinate transformation

$$x' = x - vt, \quad t' = t \qquad \Longleftrightarrow \qquad x = x' + vt', \quad t = t'. \tag{2}$$

Meaning of invariance. Equation (1) is called *Galilean invariant* if the following holds: whenever $\psi(x,t)$ is a solution of (1) in the unprimed frame, there exists a transformation rule $\psi \mapsto \psi'$ such that the transformed wavefunction $\psi'(x',t')$ satisfies the *same* Schrödinger equation written in primed coordinates,

$$\mathrm{i}\hbar\,\partial_{t'}\psi'(x',t') = -\frac{\hbar^2}{2m}\,\partial_{x'}^2\psi'(x',t'). \tag{3}$$

In short: *solutions must map to solutions under the boost*.

(i) **Failure of naive coordinate relabeling.**

A tempting but incorrect assumption is that the observer in the primed frame associates to the state the wavefunction obtained by simple coordinate relabeling,

$$\psi'(x',t') \stackrel{?}{=} \psi(x'+vt',t'). \tag{4}$$

- Use the chain rule to show that
$$\partial_{x'} = \partial_x, \qquad \partial_{t'} = \partial_t + v\,\partial_x. \tag{5}$$
- Apply $\partial_{t'}$ to the right-hand side of (4) and show that
$$\partial_{t'}\psi(x'+vt',t') = \partial_t\psi(x,t) + v\,\partial_x\psi(x,t). \tag{6}$$
- Insert (4) into (3) and show that an extra term $\mathrm{i}\hbar v\,\partial_x\psi$ appears, which cannot be cancelled.
- Conclude that the Schrödinger equation is *not* invariant under the naive assumption (4).

(ii) **Galilean invariance up to a phase.**

Define instead the transformed wavefunction

$$\psi'(x',t') = \exp\left[\frac{\mathrm{i}}{\hbar}\left(mv\,x' + \tfrac{1}{2}mv^2\,t'\right)\right]\,\psi(x'+vt',t'). \tag{7}$$

- Show that the derivatives acting on the exponential generate additional terms that cancel the convective contribution found in part (i).

- Verify explicitly that $\psi'(x', t')$ satisfies the Schrödinger equation (3).
- Explain why the phase in (7) is called *local*.

(iii) **From global to local phase transformations.**

Consider a spacetime-dependent rephasing,

$$\psi(x,t) \longrightarrow \psi_\chi(x,t) = \exp\left[\frac{\mathrm{i}}{\hbar}\, q\, \chi(x,t)\right] \psi(x,t), \tag{8}$$

with arbitrary real $\chi(x,t)$.

- Compute $\partial_t \psi_\chi$ and $\partial_x \psi_\chi$.
- Show that inserting ψ_χ into (1) produces additional terms proportional to $\partial_t \chi$ and $\partial_x \chi$.
- Conclude that the free Schrödinger equation is not invariant under local phase transformations.

(iv) **Covariant derivatives and minimal coupling.**

Introduce two real fields $\phi(x,t)$ and $A(x,t)$ and define

$$D_t := \partial_t + \frac{\mathrm{i}q}{\hbar}\, \phi(x,t), \qquad D_x := \partial_x - \frac{\mathrm{i}q}{\hbar}\, A(x,t). \tag{9}$$

- Write the gauge-covariant Schrödinger equation

$$\mathrm{i}\hbar\, D_t \psi = -\frac{\hbar^2}{2m}\, D_x^2 \psi. \tag{10}$$

- Determine how ϕ and A must transform under (8) so that $D_\mu \psi$ transforms covariantly.
- Verify that (10) is form-invariant.

(v) **Interpretation.**

The Schrödinger equation is Galilean invariant only up to a phase. Requiring invariance under *local* phase transformations forces the introduction of gauge fields, and the constant q acquires the interpretation of a charge. This provides the nonrelativistic seed of gauge theory.